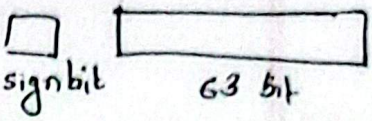
64 bit integer ⟶ largest 2^{63}



Lecture 1

Floating Point Number:

$(10.11)_2$

Left shift ⟶ $(1.011)_2 \times 2^1$

Left shift ⟶ $(0.1011)_2 \times 2^2$ ⟶ Floating point

$(0.d_1d_2d_3\ldots.d_n)_\beta \times \beta^e$ ⟶ Exponent

β ⟶ Base

10.11

10 ⟶ Whole number

11 ⟶ Fraction/ significant/ Mantissa/ Precision

Disadvantage of floating point numbers:

1. Numbers not equally spaced
2. Can only represent a finite set of numbers

$\beta = 2$, $m = 3$, $e = [-1, 2]$

$-1 \le e \le 2$

$e = [-1, 0, 1, 2]$

Ans:

$(0.___)_2 \times 2^{e=[-1,0,1,2]}$

$= (0.111)_2 \times 2^{-1}$

$= (0.101)_2 \times 2^2$

Randomly taken

$= (0.000)_2 \times 2^{-1}$ [Smallest in positive] Non negative smallest

$= -(0.111)_2 \times 2^2$ [Smallest overall]

↳ Negative support

Floating Point:

1. Standard format: $\pm(0.d_1 d_2 d_3 \ldots d_n)_\beta \times \beta^e$
   [General / Convention-1] $d_1 = 1$ & $d_1 \neq 0$

2. Normalized form: $\pm(0.1\, d_1 d_2 d_3 \ldots d_n)_B \times B^e$

3. Denormalized form: $\pm(1.d_1 d_2 d_3 \ldots d_n)_B \times \beta^e$

---

(1) Standard form:

(a) $\beta = 2$, $m = 3$, $e \in [-1, 2]$. Find the largest, smallest, non-negative smallest

Ans: Format: $(0.d_1 d_2 d_3)_2 \times 2^e$ $[d_1 = 1]$

Largest: $(0.111)_2 \times 2^2$

Smallest: $-(0.111)_2 \times 2^2$

Non negative smallest: $(0.100)_2 \times 2^{-1}$

(b) Variations:

Ans:
$(0.100)_2 \times 2^e$
$(0.101)_2 \times 2^e$
$(0.110)_2 \times 2^e$
$(0.111)_2 \times 2^e$

$4 \times 2 = 32$

4 → No of exponents; 2 → Positive and negative

Standard: $2^{m-1} \times e \times \text{sign}$

Normal/De: $2^m \times e \times \text{Sign}$

$(0.100)_2 \times 2^{-1}$ → Decimal → $\frac{4}{8} \times \frac{1}{2} = \frac{1}{4}$

Considering positive numbers only.

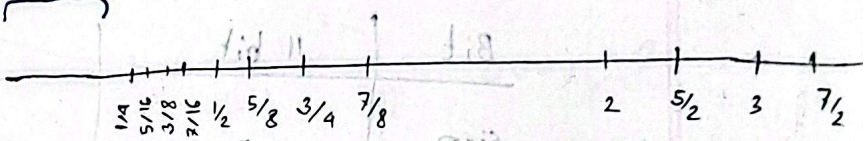
$(0.100)_2 \times 2^{-1} = \frac{1}{4}$

$(0.101)_2 \times 2^{-1} = \frac{5}{16}$ — 1/16

$(0.110)_2 \times 2^{-1} = \frac{3}{8}$

$(0.111)_2 \times 2^{-1} = \frac{7}{16}$ — 1/16

Consecutive difference



→ e ছোট তাই consecutive difference কম

→ Range ও কম

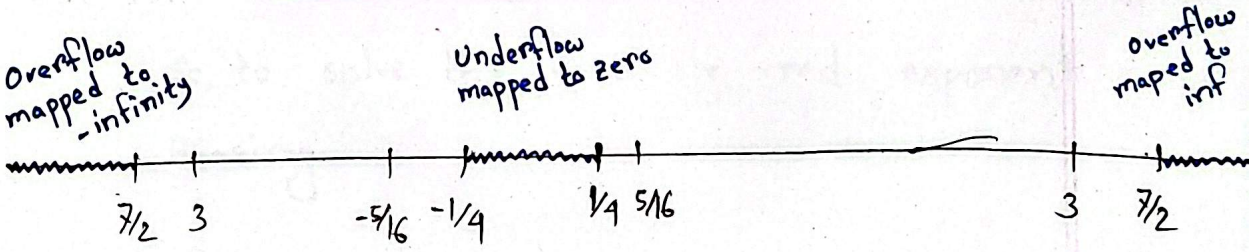
→ e বড় তাই consecutive difference বেশি

→ Range বেশি

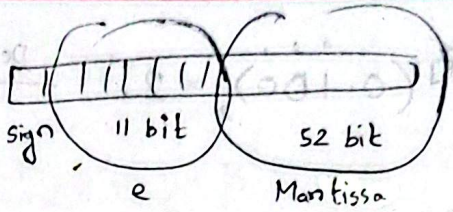
* If exponent is fixed ~~and mantissa is~~ fixed, underflow and overflow occurs.

[They are not equally spaced]

Considering all positive and negative numbers



IEEE standard (64 Bit):



Bit	11 bit	52 Mantissa
Sign	exp	
0 (-ve)	e	
1 (+ve)		

Minimum exponent = 0

Maximum exponent = 2047

$e = [0, 2047]$

e_{min}

$e_{max} = 2^{11} - 1$

② Normalized form:

$B = 2, m = 3, e \in [-1, 2]$

Ans: $(0.1\text{---})_2 \times 2^e$

- Largest: $(0.1111)_2 \times 2^2$
- Non negative smallest: $(0.1000)_2 \times 2^{-1}$

③ Denormalized:

$B = 2, m = 3, e \in [-1, 2]$

Ans: $(1.\text{---})_2 \times 2^e$

- Largest: $(1.111)_2 \times 2^2$
- Non negative smallest: $(1.000)_2 \times 2^{-1}$

IEE format: $(1.\underbrace{111111\ldots}_{52})_2 \times 2^{2047}$ [Largest Denormalized] = [X] (Too big to use)

$\therefore (1.\underbrace{0000\ldots}_{52})_2 \times 2^{0}$ [Smallest Denormalized = 1]

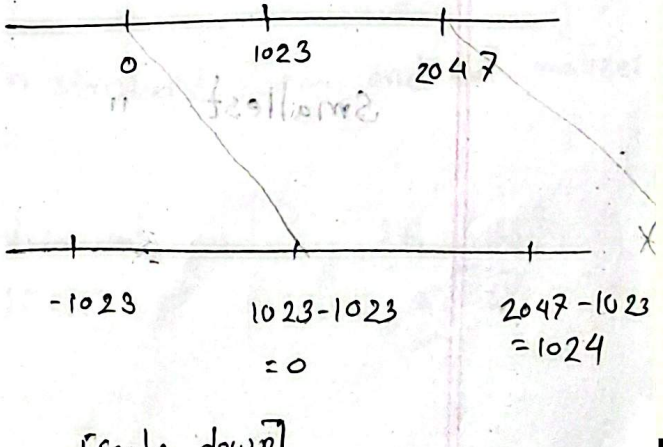
~~exponen~~ (Can't express smaller numbers)

So, to solve the issue we need exponent Biasing.

$e \in [0, 2047]$ → Exponent:

$\frac{2047}{2} = 1023$

Exponent Biasing:



$e \in [-1023, 1024]$

[Scale down]

Denormalized form in exponent biasing:

$$\pm (1 \cdot d_1 d_2 \ldots\ldots d_{52})_\beta \times \beta^{e-1023}$$

(Normalized form) $\pm (0.1 d_1 d_2 d_3 \ldots d_{51})_\beta \times \beta^{e-1023+1}$

CamScanner

Lecture 2

Rounding Error

→ In number line we can only present discrete set of numbers as floating points

$\pi = 3.1415926535.....$ [Actual value gives more accuracy]

We use,

$\pi = 3.14$ [Actual value এর certain portion]

$$\text{Error} = |\pi - fl(\pi)| \quad \text{(Actual value - Calculated value)}$$

$$= 3.1415926535.... - 3.14$$

$$= 0.005926.....$$

Safe

1000

↳ যদি 0.1 error

Dangerous

0.5

↳ যদি 0.1 error

Scale invarient rounding error, δ :

Relative rounding error

$$\delta = \frac{|fl(x) - x|}{|x|}$$

$fl(x)$ = calculated value

x = Actual value

$x = 3$ $fl(x) = 2.5$. $\delta = ?$

$$\delta = \frac{|2.5 - 3|}{3}$$

→ Always Decimal

$$|fl(x) - x| \leq \frac{1}{2}\beta^{-m}\beta^{e}$$

* $(6.25)_{10}$. Find (a) $fl(6.25)_{10}$ In standard form and denormalized $\beta = 2, m = 3, [-1 \leq e \leq 2]$ (b) δ

(a) $(0.11001)_2 \times 2^3$

[It is overflow]

∴ standard

~~If $e = 3$~~, e_{max},

∉

In denormal,

$(1.1001)_2 \times 2^3$

$= (1.100)_2 \times 2^2$

$= 6$

(b) $\delta = \frac{|6 - 6.25|}{6.25}$

1.449
1.45

Maximum Scale invarient R.E. / Machine Epsilon / Upper bound: Max. Delta / Unit Roundoff:

$$\epsilon_{max} / \delta_{max} = \frac{|fl(x) - x|}{|x|}$$



Prepare for worst case

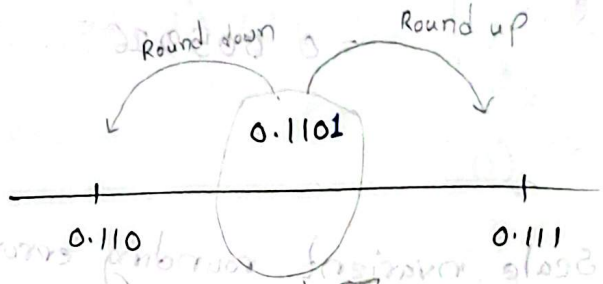
Now,

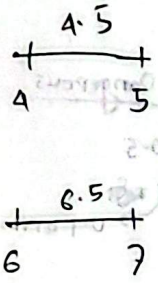
$$\delta_{max} \propto |fl(x) - x| \quad \& \quad \delta_{max} \propto \frac{1}{|x|}$$

~~Derivation~~

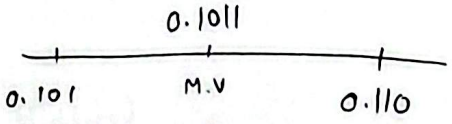
* $1.449 \xrightarrow{\text{Rounded}} 1.45$

m = 3, standard form:



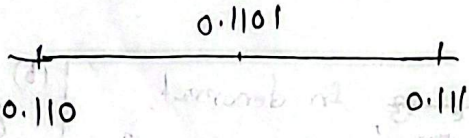


$0.110 + 1 \rightarrow 0.111$



0.110101, m = 3

Ans: 0.111

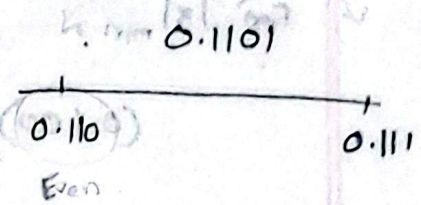


mid পর্যন্ত আসলে এখানে ক্লিক করে আবার শুরু থেকে দেখবে

Mid পয়েন্ট বা মিলে বসা হবে

0.11100101 . m=3

Ans: 0.110 [left]



(*) 0.1101 . m=3

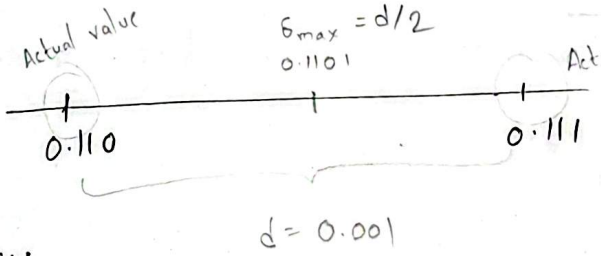
Ans: 0.110 [Even]

(Exactly mid (0.1101) হলে, যেটা even (LSB = 0) সেটা হবে)

Standard form - Machine Epsilon, δ_{max}, m=3, β=2

Ans: $(0.d_1 d_2 d_3)_\beta \times \beta^e$

Maximum value of delta



$$\begin{array}{r} 0.111 \\ -\ 0.110 \\ \hline 0.001 \end{array}$$

Denominator:

$$|fl(x) - x|_{max}$$

$$= \left((0.001)_2 \times 2^e\right) \times \frac{1}{2}$$

$$= \left((0\times 2^{-1} + 0\times 2^{-2} + 1\times 2^{-3})_2 \times 2^e\right) \times \frac{1}{2}$$

↓ Half convert হবে না

$$= \left(2^{-3} \times 2^e\right) \times \frac{1}{2}$$

$$= \left(\beta^{-m} \times \beta^e\right) \times \frac{1}{2}$$

For $|x|_{min}$,

$$(0.100)_2 \times 2^e$$

$$= 1 \times 2^{-1} \times 2^e$$

$$= 2^{-1} \times 2^e$$

$$= \beta^{-1} \beta^e$$

So,

$$\delta_{max} = \frac{|fl(x) - x|}{|x|}$$

$$= \frac{\frac{1}{2} \beta^{-m} \beta^e}{\beta^{-1} \beta^e}$$

$$= \frac{1}{2} \beta^{-m+1}$$

In normalized form:

$$|fl(x) - x| = \beta^{-m-1}\beta^{e}$$

$$|x|_{min} = (0.1000) \times 2^{e}$$

$$= 2^{-1}2^{e}$$

$$= \beta^{-1}\beta^{e}$$

$$\delta_{max} = \frac{\frac{1}{2}\beta^{-m-1}\beta^{e}}{\beta^{-1}\beta^{e}}$$

$$= \frac{1}{2}\beta^{-m}$$

Denormalized form:

$$|fl(x) - x|_{max} = \beta^{-m}\beta^{e}$$

$$|x|_{min} = (1.000)_2 \times 2^{e}$$

$$= 2^{0} \times 2^{e}$$

$$= 2^{e}$$

$$= \beta^{e}$$

$$\delta_{max} = \frac{\frac{1}{2}\beta^{-m}\beta^{e}}{\beta^{e}}$$

$$= \frac{1}{2}\beta^{-m}$$

Sunday 21.06.26

Problems of rounding error:

$x = \frac{5}{8}$

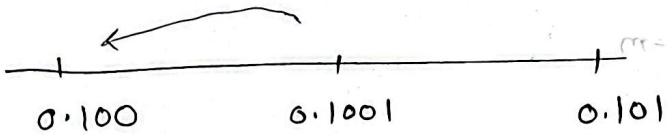
$y = \frac{7}{8}$

$xy = \frac{35}{64} = 0.5468$ [Actual answer]

$xy = \frac{5}{8} \times \frac{7}{8}$

$= (0.100011)_2$

$\beta = 2$, $m = 3$ conversion of xy



$xy \approx (0.100)_2$

$= (0.5)_{10}$

Lost of Significant

* Left থেকে right
* Non zero থেকে নেওয়া শুরু

#3 significant figure / value

① $10.110 \approx 10.1$

② $0.00100011 \approx 0.100 \times 2^{-2}$

Mantissa bit কমায় দিয়ে exponent bit বাড়ায় দিবো

③ $0.11011 \approx 0.1101$

④ $0.11 \approx 0.110$

Example : 4 significant figure

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$x^2 - 56x + 1 = 0$

$\Rightarrow x_1 = 28 + \sqrt{783} = 55.98$ [Actual answer]

$\Rightarrow x_2 = 28 - \sqrt{783} = 0.01786$ [Actual answer]

$x^2 - (\alpha+\beta)x + \alpha\beta$

$\Rightarrow \alpha = x_1$

$\Rightarrow \beta = \frac{1}{x_1} = x_2$

$\sqrt{783}$

$= 27.982$

$= \boxed{27.98}$

Loss of Significance

Quadratic equation is $x^2 - 60x + 1 = 0$. (a) Working to 6 significant figures, compute the roots of quadratic equation and (b) check that there is loss of significance. (c) Find the correct root such that loss of significance does not occur.

Ans: $x = \frac{-(-60) \pm \sqrt{(-60)^2 - 4 \times 1 \times 1}}{2 \times 1} = 30 \pm 29.9833$

$= 59.9833$ and

0.0167000 (6 s.f)

But the product of roots must be equal to 1, but we get $59.9833 \times 0.0167000 = 1.00172 \neq 1$. There is loss of significance due to subtraction of two close numbers 30 and 29.9833.

One root is $= 59.9833$

So, $x_2 = \frac{1}{x_1} = \frac{1}{59.9833} = 0.0166713$ up to 6 s.f

Now, check $59.9833 + 0.0166713 = 59.99997$

$= 60.0000$ (6 s.f)

Consider quadratic eqn $x^2 - 16x + 3 = 0$. Explain how the loss of significance occur in finding roots of quadratic eqn if we restrict to 4 s.f. Discuss how to avoid this.

Ans:

$$x^2 - 16x + 3 = 0$$

$$x = \frac{16 \pm \sqrt{(-16)^2 - 12}}{2} = 8 \pm \sqrt{61}$$

$$= 8 \pm 7.810$$

$$= 15.81 \text{ and } 0.19$$

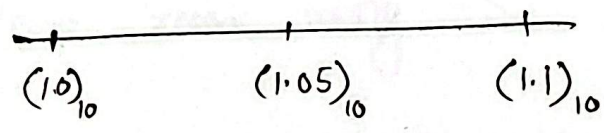
$$\therefore 15.81 \times 0.19 = 3.004 \neq 3$$

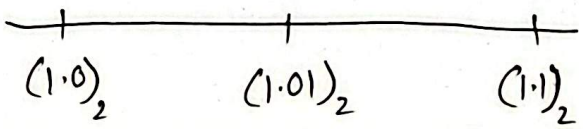
$$2^{nd} \text{ root} = \frac{3}{15.81} = 0.1898 \ (4 \text{ sf})$$

Why even base β is required for representing numbers in floating point representation.

Ans:

In even base (e.g., $\beta = 2, 10, 16$) mid point between two consecutive numbers is a finite digit sequence





In odd base (e.g., $\beta = 3, 5$), the mid point can not be represented with a finite digit sequence

